

ZAYAN AHMAD GHOUS

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GitHub: github.com/ZayanAhmadGhous | Portfolio: zayanahmad.me

PROFESSIONAL SUMMARY

BS Computer Science student with hands-on experience in Machine Learning, Computer Vision, and AI application development. Built end-to-end ML pipelines, REST APIs using FastAPI, and deployed AI applications with TensorFlow, Docker, and Hugging Face Spaces. Passionate about building scalable AI solutions and contributing to production-ready machine learning systems.

TECHNICAL SKILLS

- Programming Languages:** Python, C++, C, SQL
- Machine Learning & AI:** TensorFlow, Keras, scikit-learn, Convolutional Neural Networks (CNN), model optimization, cross-validation
- Data Science:** Pandas, NumPy, Matplotlib, Feature Engineering, Exploratory Data Analysis (EDA), Data Processing
- Backend:** FastAPI, Django REST Framework, REST API design, API testing (Postman), Gradio
- Tools:** Git, GitHub, Docker, Jupyter Notebook, Google Colab, Hugging Face Spaces, Streamlit, Linux terminal
- Foundational Knowledge:** Transformer Architecture, Large Language Models (LLMs)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern

University of Engineering & Technology (UET) Lahore

May 2025 – July 2025

- Developed and trained multiple machine learning models including Decision Trees, Random Forests, and Neural Networks using Python and scikit-learn, achieving 85%+ accuracy on real-world datasets
- Built end-to-end data preprocessing pipelines including missing value handling, feature scaling, categorical encoding, and data transformation for datasets with 10,000+ samples
- Worked on data engineering tasks including data cleaning, ETL workflows, dataset integration, and automated preprocessing pipelines using Pandas and NumPy
- Designed CNN-based image classification models for MRI brain scan analysis using TensorFlow/Keras with data augmentation techniques to improve model robustness
- Deployed machine learning applications using FastAPI and Gradio for real-time inference and interactive web-based predictions
- Performed exploratory data analysis (EDA), feature engineering, and model evaluation while managing projects with Git/GitHub and Jupyter Notebook

PROJECTS

Brain Tumor Detection (CNN + FastAPI + Deployment)

Live: huggingface.co/spaces/Ghous/brain_tumor

- Built a CNN-based brain tumor classification model achieving 88% accuracy on MRI dataset (500+ images) using TensorFlow/Keras
- Improved model generalization using data augmentation, dropout regularization, batch normalization, and early stopping
- Developed REST API using FastAPI for real-time inference and integrated with a Gradio UI for user-friendly predictions
- Deployed end-to-end ML application on Hugging Face Spaces for public web access
- Tech Stack: Python, TensorFlow, Keras, FastAPI, Gradio, scikit-learn

House Price Prediction API (End-to-End MLOps Pipeline)

Docker Hub: [zayanahmad123/house-price-api](https://github.com/zayanahmad123/house-price-api)

- Built a house price prediction model using scikit-learn with feature engineering, outlier removal, and categorical encoding on a real-world housing dataset
- Developed automated preprocessing pipelines using Pandas, SciPy, and MEstimateEncoder for reliable model training and inference
- Created a FastAPI-based REST API for real-time house price predictions with production-ready inference workflows
- Containerized the complete application using Docker and published the image on Docker Hub for reproducible deployment
- Tech Stack: Python, scikit-learn, Pandas, NumPy, FastAPI, Docker, SciPy

EDUCATION

Board of Intermediate and Secondary Education

Intermediate in Computer Sciences (Punjab Group of colleges)

Lahore, Pakistan

2022 – 2024

Minhaj University Lahore

BS in Computer Sciences | CGPA (3.66 / 4.00)

Lahore, Pakistan

2024 – Present